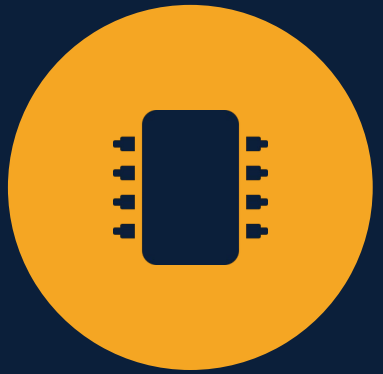
UNIT 3 · 9 hours

Half-Wave Rectifier



Diodes, Transistors and their Applications

Basic Electrical and Electronics Engineering (EEE103)
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NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

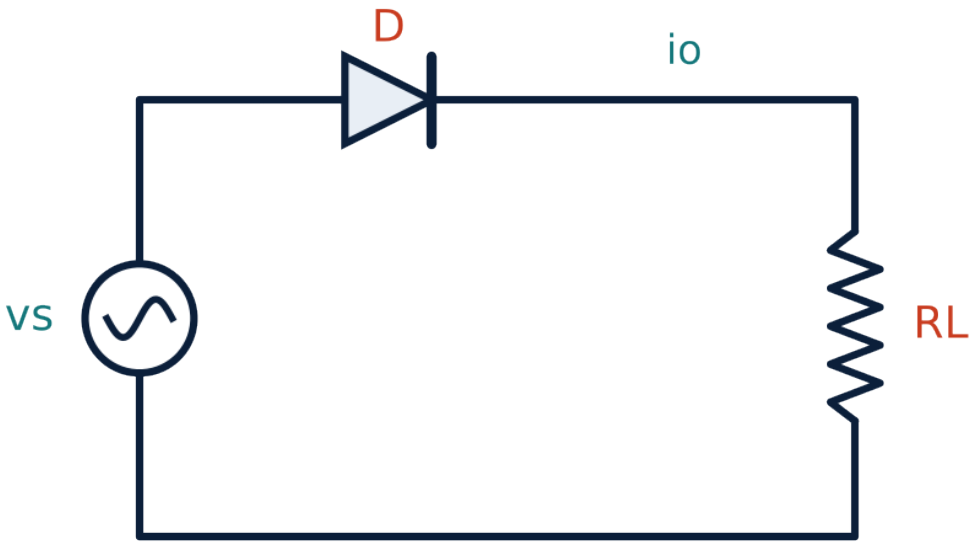
- Explain the operation of the half-wave rectifier
- Derive Vdc, Idc, Irms, ripple factor and efficiency
- State the peak inverse voltage
- Solve numericals

Syllabus context

Semiconductor diode, diode equivalent circuits, half-wave rectifier, full-wave bridge rectifier. Zener diode and its use in voltage regulation.

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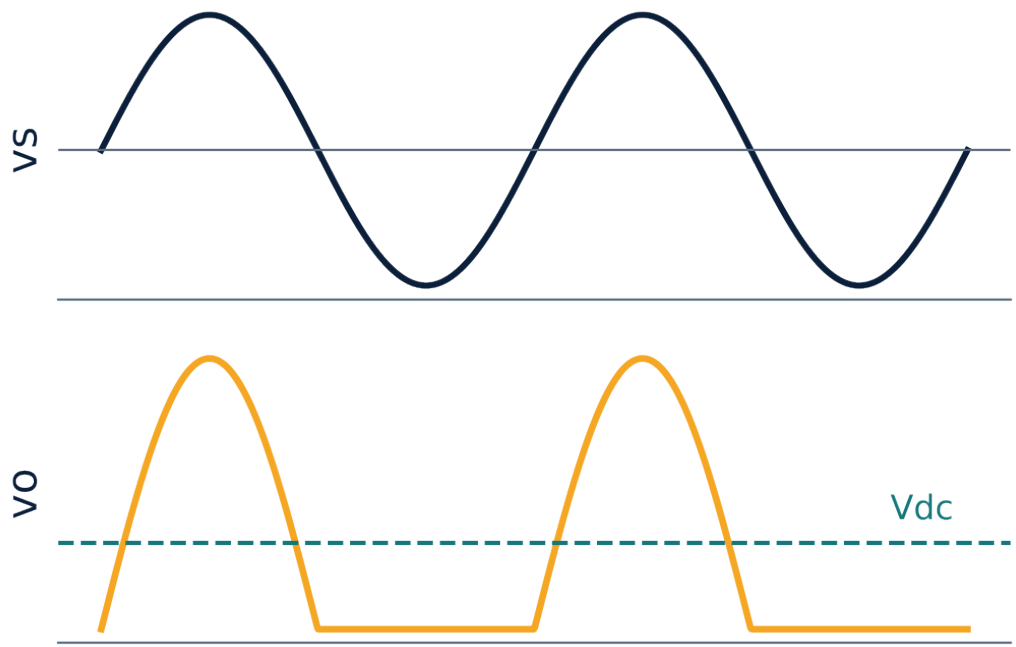
Circuit and operation



- Single diode D in series with load RL.
- Positive half-cycle: D forward biased, current flows through RL.
- Negative half-cycle: D reverse biased, no current.
- Output is a pulsating DC with only positive half-cycles.

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Input and output waveforms



- vo = Vm sin θ for 0 < θ < π, zero otherwise.
- Output frequency equals input frequency.
- Dashed line: average (DC) value Vdc = Vm/π.

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Performance equations (Rf = diode resistance)

$$I_m = V_m / (R_f + R_L)$$

peak current

$$I_{dc} = I_m / \pi$$

average current

$$I_{rms} = I_m / 2$$

RMS current

$$V_{dc} = I_{dc} R_L = I_m R_L / \pi$$

DC output voltage

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Efficiency, ripple and PIV

$$\eta = P_{dc}/P_{ac} = 40.6\ \%$$

maximum (Rf = 0)

$$\gamma = \sqrt{((I_{rms}/I_{dc})^2 - 1)} = 1.21$$

ripple factor

$$PIV = V_m$$

peak inverse voltage

- $P_{dc} = I_{dc}^2 R_L$; $P_{ac} = I_{rms}^2 (R_f + R_L)$.
- Ripple factor 1.21 means ripple exceeds the DC value: poor DC quality.
- Transformer utilisation factor is only 0.287.

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Numerical: half-wave rectifier

Problem Secondary voltage 20 V RMS, 50 Hz, RL = 100 Ω, diode ideal (Rf = 0). Find peak, DC and RMS current, power and efficiency.

- $V_m = 20\sqrt{2} = 28.28$ V; $I_m = 28.28/100 = 0.283$ A
- $I_{dc} = I_m/\pi = 90.0$ mA; $I_{rms} = I_m/2 = 141.4$ mA
- $P_{dc} = I_{dc}^2 R_L = 0.81$ W; $P_{ac} = I_{rms}^2 R_L = 2.0$ W
- $\eta = 0.81/2.0$

Answer: I_{dc} = 90 mA, $\eta \approx 40.5$ %

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Applications and limitations

Used in: low-power circuits, simple battery chargers, AM demodulation, peak detectors.

Drawbacks: high ripple, low efficiency, DC component saturates the transformer core.

Improvement: add a filter capacitor, or use a full-wave rectifier.

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Key takeaways

- Conducts only during positive half-cycles
- Vdc = Vm/π, Idc = Im/π, Irms = Im/2
- η max = 40.6 %, ripple factor = 1.21
- PIV = Vm

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Practice questions

Q1. Derive Idc and Irms for a half-wave rectifier.

Q2. Vm = 50 V, RL = 1 kΩ, Rf = 0. Find Vdc and Irms.

Q3. Why is the ripple factor high for a half-wave rectifier?

Q4. What is the PIV rating needed for 230 V RMS input?